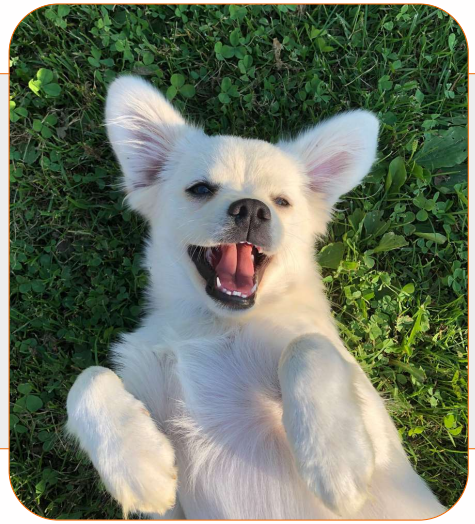
DOG BREED EXPLORER



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LT2319

Introduction



You can tell it, by voice, what kind of dog would fit your life. It narrows down breeds and answers your questions about them.

1.Ask

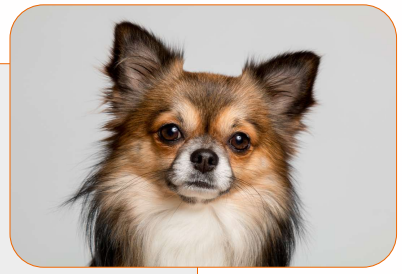
Elicits size, activity level, grooming and temperament, one question at a time, or all at once.

2.Recommend

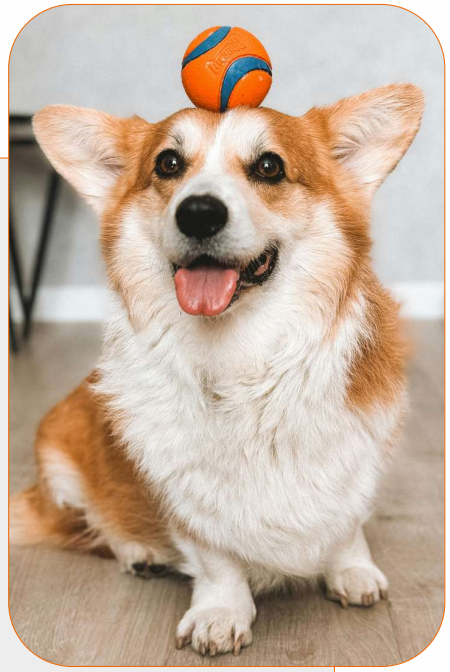
Matches breeds from TheDogAPI and says why each one fits, three at a time.

3.Explain

Answers follow-up questions (“does it shed a lot?”) using retrieved breed texts.



Pre-existing Services and APIs



Service	Role in DogExplorer
TheDogAPI	Breed database: weight, height, temperament, breed group, life span, description
SpeechState + Azure	Speech recognition and synthesis in the browser (en-US)
Ollama.llama3.1	NLU to JSON, RAG answer generation, free replies to off-topic turns
Ollama.qwen3-embedding	384-dimensional embeddings of breed texts and questions
Qdrant	Vector database for retrieval, filtered by breed
XState	State machine for the spoken turn loop

Scenario

**A person who wants a dog, but doesn't know which breed, describes what they need.
The system fills in what is missing and suggests breeds.**

User: "I'm looking for a small, friendly dog that doesn't need much grooming."

System: "How active would you like the dog to be?"

Size, temperament and grooming are given → only activity is still missing, so that is what the system asks.

Collected material

Source	Content	Used for	Amount
TheDogAPI	Structured attributes, description, history	Matching + RAG	630 breeds
Wikipedia	Care sections: temperament, health, grooming, exercise	RAG	59 breeds
Test set	23 breeds saved from TheDogAPI + scripted user utterances	Scenario tests	23 breeds

Example: Beagle

TheDogAPI

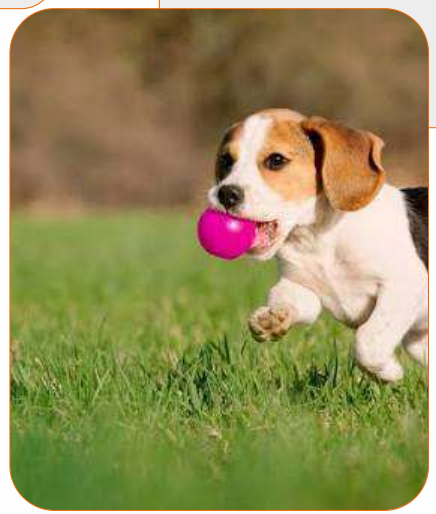
Temperament: Friendly, gentle, alert, outgoing, curious, energetic
Breed group: Hound
Life span: 10–15 years
Weight: 9–14 kg

Wikipedia

Temperament:
The Beagle has an even temper and gentle disposition. Described in several breed standards as “merry”… Beagles are excellent with children.

Wikipedia

Health:
A 2024 study in the UK found a life expectancy of 12.5 years… Beagles may be prone to epilepsy, but this can often be controlled with medication.



The API gives facts for matching; the texts answer “how” and “why” questions.

Sceniario to a Dialogue Design

User intents

set_preferences
show_results
more_options
select_result
ask_about_breed
general_question
restart · help

Preference slots

size: small, medium, large
activity: low, medium, high, any
grooming: low, medium, high, any
temperament: friendly, calm…
Operation: replace or add

Breed questions

description , size
temperament, lifespan
grooming, exercise
apartment, training
health, other

TheDogAPI has no activity or grooming fields

Users ask about exactly these two things, so the matcher estimates them from the words in each breed's text.

Trait	Estimated from	Example signals
Size	Metric weight range	average weight → small / medium / large
Activity	Temperament, description, breed group	"energetic", "athletic" ↑ · "calm", "lap dog" ↓ · Sporting, Herding ↑ · Toy ↓
Grooming	Coat words in the description	"long", "silky", "curly" ↑ · "short", "smooth" ↓ · "double", "wiry" = medium

The estimate is also used to explain each result: "Miniature Schnauzer, small, very active, moderate grooming."

Architecture

Speech in

Azure ASR via SpeechState

→

NLU

Fixed phrases first, then llama3.1 → JSON

→

Update rules

Integrate → Execute → Select, on the information state

→

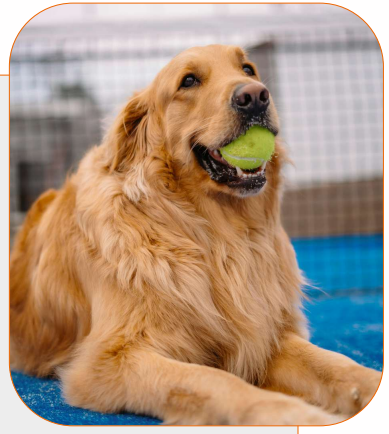
NLG

System moves → text

→

Speech out

Azure TTS



The Execute Step consults:

TheDogAPI

Search and rank breeds; size and life span answers

RAG (Qdrant)

Grooming, exercise, health, training questions

LLM fallback

Small talk and open questions; else repair

Retrieval for breed question

Indexing

1. Fetch TheDogAPI and Wikipedia texts
2. Split into 500-character chunks, 50 overlap
3. Embed with qwen3-embedding (384 dims)
4. Store in Qdrant with breed + source URL

Answering (each question)

1. Embed the user's question
2. Top 4 chunks, filtered to the current breed
3. llama3.1 answers from those chunks only
4. No answer in the text → "I don't have that information"

What it handles , hard parts

Dialogues it can handle

- Several preferences in one utterance
- Corrections that replace or add
- “I don’t mind” for the slot just asked
- “The second one”, “it”, “anything else?”
- Breed questions from the API or RAG
- Small talk via the LLM; repair, help, restart

Problems encountered

- No activity/grooming in the API: word-based estimates can be wrong (“long ears” ≠ long coat)
- LLM NLU is slow and inconsistent on short answers: fixed phrases + last question as context
- RAG could invent answers: NO_ANSWER instead

What would improve the system

Test with real users

Record real dialogues and see where the NLU and the questions fail.

Better trait data

Use rated energy and grooming levels instead of word estimates.

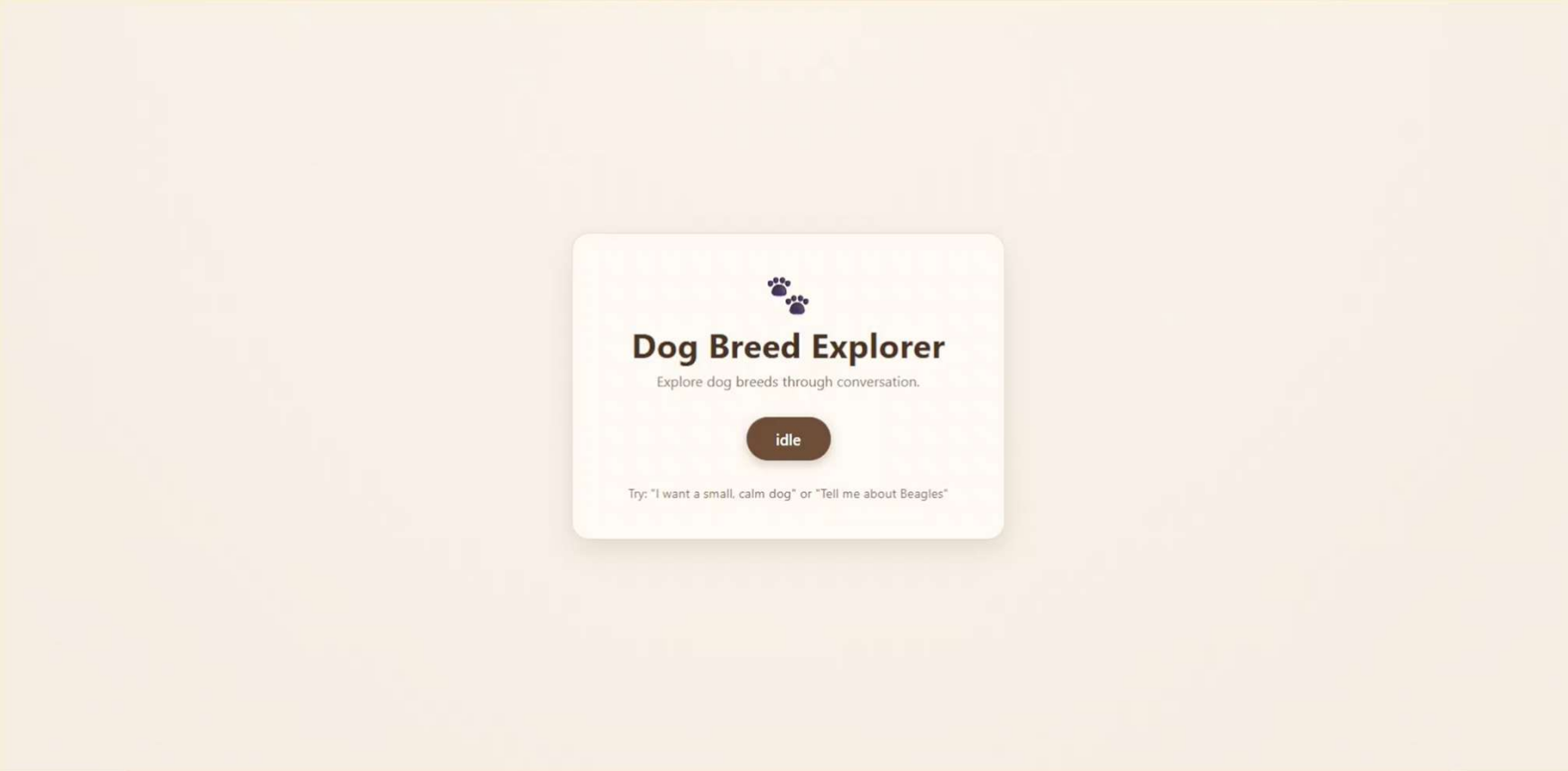
Compare breeds

“Which is calmer, the Beagle or the Whippet?”

More slots and more texts

Children, other pets, Wikipedia for more breeds.

Demo


Dog Breed Explorer
Explore dog breeds through conversation.
idle
Try: "I want a small, calm dog" or "Tell me about Beagles"

Thank You!